On my honor, I have neither given nor received any unauthorized and inappropriate assistance for all sessions of this exam. The work done on this exam is to tally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh disciplin issues

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A)

1) $kx_2 + x_4 = 0$ $\quad -\frac{1}{2}x_3 + x_4 = 0$

$x_2 = x_3$

$mx_1 + x_3 = 0$ $\quad -8x_1 + x_2 = 0$

$mx_1 = -x_3$ $\quad x_2 = 8x_1$

$x_2 = x_3$ and $x_2 = 8x_1$ ve $-mx_1 = x_3$

$-mx_1 = 8x_1$

$m = -8$ $\quad k = -\frac{1}{2}$

c)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -\frac{1}{2} & 0 & 1 \end{bmatrix} = 1 \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 0 & 1 & 0 \\ -8 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} = 1 \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{bmatrix} = 1$$

3)

a) $H^2 = H \cdot H = (I - 2uu^t)(I - 2uu^t)$

$= I^2 - I \cdot 2uu^t - 2uu^t \cdot I + 4uu^t \cdot uu^t$

$= I(I - 2uu^t - 2uu^t) + 4u \cdot u^2 \cdot u^t$

$= I - 2uu^T - 2uu^t + uu \cdot u^2 \cdot u^t$

P's $\|u\|^2 = u^t \cdot u$

1 e $\|u\| = \pm\sqrt{u^s u}$ $\quad \|u\| = 1$

$H^2 = I - 4uu^t + uuu^s$

$H^2 = I - 4(1) + 4(12$

$H^2 = I$

b) $HH^s = (I - 2uu^s)(I - 2uu^t)^t$

$= (I - 2uu^t) \cdot (I^T - (2uu^t)^t)$ $\quad (\because (A - B)^s = A^T - B^t$

$= (I - 2uu^t) \cdot (I - 2(u^t)^t u^s)$

$= (I - 2uu^t) \cdot (I - 2uu^s)$ $\quad [\because (AB)^t = B^t \cdot A^t$

$= I^2 - 2uu^t - 2uu^t + 4uu^t \cdot uu^T$

$= I - 4uu^t + 4u\|u\|^2 u^T$ $\quad (\because A \cdot (u^t)^t = u.$

$= I - 4uu^t + 4uu^t$ $\quad (\because \|u\| = 1)$

$= I - 4(1) + 4(2)$

$H \cdot H^t = I$

$H^t H = I$ $\quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

so, $HH^T = H^t H = I$

so, H is on orthogonalmatrix

c) $Hu = (I - 2uu^t) \cdot u$

$= Iu - 2uu^s u$

$= u - 2u - \|u\|^2$ [As I is identify matrix as,

is a Identify matrix, so is the only nisen value

$Iu = 1 \cdot u = u$ also, and $\|u\| = +\sqrt{u^s u}$

$\|u\| = 1 = +\sqrt{u^t u}$

$u^s u = \|u\|^2 = 1$

$Hu = u - 2u \cdot 1 = -u$

$Hu = -u = (-1)\, u.$

we have → $Hu = \lambda u$ hence $\lambda = -1$

$u \bar{u} = \lambda = -1.$

d) given, $H = (I - 2uu^t)$

as, $H^t = (I - 2uu^t)^t$

$= I^t - 2(uu^t)^t$ (∵ $(A-B)^t = A^t - B^t$

$= I - 2(u^t)^t u^t$ ∵ $((AB)^t = B^t \cdot A^t)$

$= I - 2uu^t = H$

H is symmetric matrix H is diagonalizable matrix

$H = Q \Lambda Q^s$

$= Q \Lambda Q^{-1}$

$= (Q^{-1})^{-1} \Lambda Q^{-1}$

$P^{-1} \Lambda B$